

The Case for IT Governance in Retail Supply Chains: An Overview

ACC305 Team A5

Bella Hughes, Joseph Sher, Zach Villeneuve

Professor Jeffrey Merhout

Thursday, March 19, 2026

Honor Pledge:

“On my honor, I have not committed plagiarism, neither explicitly or inadvertently, on this project.”

AI Use: Our team sparingly used AI, only helping us to compile an outline from our notes and understand the best structure to use when building this framework and organizing the project. We used it to help us compile our long list of notes into a more digestible summary that led us to pull key insights from the notes and elaborate on key risks and concerns, with all writing, analysis, and framework design done by the team.

Table of Contents:

Executive Summary	3
Thesis	3
Introduction	4
IT Governance Definition	4-5
Key Risks	5-8
Framework	8-11
Conclusion	12
References	13-23

Executive Summary:

This report examines the importance of a sound IT governance framework to protect big data within retail supply chains. We, as internal auditors for BJZ Corporation, were tasked to propose a comprehensive governance framework for BJZ's upper management, which is responsible for strategic oversight and use of big data across supply chain functions. Throughout this report, we develop an IT governance definition, identify nine risks regarding retail supply chain data, and propose an IT governance framework containing six domains that will help guide retailers to manage and protect their data within their supply chains.

Beginning with our IT governance definition, IT governance is about specifying an organization's decision rights and accountabilities to encourage strategic alignment regarding the goals of businesses while managing risk and complying with regulations. By following this definition, retail supply chains in organizations will be able to benefit all their stakeholders and achieve any organizational goals or objectives.

Next, in this report we identify nine risks associated with retail supply chains. These risks are data quality, consumer privacy, third-party risk, incentive misalignment, cybersecurity, bias in analytics, overreliance on analytics, integration risk, and downtime & unavailability. Our framework will help mitigate these risks so that retail supply chains can uphold their integrity.

Lastly, we propose a six-domain Retail Big Data and Analytics (RBDA) Framework that will best mitigate the risks mentioned in this report. Our framework pulls together established guidelines such as COBIT, Weill (2004), COSO, and AICPA Criteria. The six domains consist of data privacy & regulatory compliance, strategic alignment, data quality & reliability, system availability & reliability, data governance & security, and managing vendor risks. **Overall, retail companies must develop an IT governance framework to address big data and analytics in their supply chains and logistics focusing on key issues such as data privacy, integration complexities, data and decision qualities, incentives, and maintaining system availability.**

Introduction:

Recent transformations to retail supply chains have created many opportunities for big data-driven decision making and enhanced operational efficiency. Large volumes of data are being collected from multiple sources along the supply chain at rapid rates, spanning from production, inventory management, consumer demand, and selecting vendors (Niu et al., 2021). The growth of big data opens the door for widespread adoption of analytics-driven systems to serve retail organizations in supply chain functions.

Traditionally, firms have depended on human intuition and managerial experience to develop meaningful insights and create value to support firm operations. Now, the current state of retail supply chains requires analytics-based models to drive effective operations and decision-making. Finding the correct balance between big data and human involvement in developing solutions drives value creation and ensures organizational goals are met.

Yet, however beneficial big data has become in the retail industry, failure to conduct proper governance and oversight of its use diminishes the same value it purports to create.

Organizations must consider who makes and is held accountable for decisions, how to maintain security of sensitive information, and how to address risks associated with big data. The lack of strong governance structures positions firms to be at risk of inaccurate or biased decisions, breach of consumer or vendor data, and ultimately, clouded judgment in reaching company strategy.

It is for these reasons that a governance framework for big data in retail supply chains is critical to help firms achieve maximum value while balancing compliance, risk management, and accountability.

IT Governance Definition:

From our research, we created a comprehensive IT governance framework definition covering important aspects such as decision rights, delivering organizational value, and ensuring proper controls to manage risks. Big data and analytical systems must be properly secure and governed,

given the large volume and value it brings. Understanding the benefits of big data analytics, we recognize that good governance requires aligning IT incentives with organizational objectives, to deliver optimal value for stakeholders and the company, while maintaining and monitoring proper controls, to ensure data integrity for making effective decisions. More formally, our definition of IT governance is:

IT governance is the organizational mechanism of “specifying decision rights and accountabilities” (Weill, 2004) that encourages “the ability for IT to achieve business goals” (Curtis, 2020) while balancing compliance and risk management to ensure “stakeholder needs, conditions, and options are evaluated” (Bakshi, 2016), especially in the context of big data in retail supply chains, where data must be secure, high quality, and reliable to be used in accordance with organizational objectives, benefit stakeholders, and deliver business value.

Key Risks:

Data Quality: Big data presents a risk of poor quality if not properly governed. With the large volumes and varieties collected by retailers amongst all activities in the supply chain, ensuring data quality is key to protecting company operations and decisions. For example, retailers rely heavily on big data analytics for forecasting, inventory management, establishing customer relationships, and choosing vendors (Brau et al., 2024). Poor data may cause a firm to overstock or understock, miscount inventory, and adversely pick an incorrect vendor. Maintaining strong oversight of data and its quality protects companies from harmful decision making and potential adverse consequences.

Consumer Privacy: Customers are a critical component of today’s retail supply chains, as retailers deliver finished goods and services directly to consumers. Each transaction, loyalty program, or promotion a customer participates in generates data that may contain sensitive information such as credit card information, address, and phone number. Retailers need to develop, implement, and establish assurance practices to confirm consumer privacy is in their best interest (Haessner et al., 2025). Breach of sensitive information can have further consequences on organizations, such as fines, legal fees, and possible legal liability for

executives. Without effective oversight, big data faces the risk of invasion of data privacy, which would affect both retailers and consumers.

Third-Party Risks: Using big data analytics typically requires cooperation with external suppliers to process and store data. Relying on a third-party to handle data is a risk since companies need to select vendors with strong controls and processes in place to protect sensitive information. In addition, retailers share customer and operational data with third-party service providers, so putting data in the hands of these suppliers introduces uncertainty (Johnson et al., 2024). Strategically choosing vendors that deliver value to the organization and to stakeholders is a governance perspective that must be considered with big data.

Incentive Misalignment: Along the different parts the supply chain, many parties use data and analytics to advance functional objectives and department goals. However, instances may occur where department goals may not align with organizational objectives, creating different incentives in using data for decision making. Internal stakeholder incentives must be aligned with the business so that value creation is met through analytics (Brau et al., 2024). When incentives are not aligned, entities can use data in a way that benefits their own good yet diminishes the value for the firm as a whole. Therefore, examining incentive alignment for big data use can prevent harmful decisions from occurring.

Cybersecurity: Scanners, sensors, POS systems, and many other technologies are present throughout the many retail supply chain stages. Each of these technologies collect large volumes of data in their operations, so there is a need to secure systems from breach of sensitive information. Cybersecurity risks in retail settings are evident in the form of accessing organizational data and altering it, stealing it, or removing or creating fraudulent transactions which could bring the company losses. In addition, access to production information such as trade secrets can be at risk of unauthorized use if not properly handled (Ardhaninggar & Ramli, 2024). Adhering to applicable cybersecurity frameworks and standards is important to reduce security risks.

Bias: Optimizing quality and integrity of information is essential to drive good decision making and establish proper conduct within an organization. The sheer volume and complexity of data retail supply chain systems collect daily needs to be complemented with effective oversight so that no data is poor in quality. Systems may be trained to follow certain algorithms in tracking patterns which could lead to bias in analytical models, such as forecasting bias and data representation bias (Johnson et al., 2024). Biased data results in minimized efficiency, inaccurate inventory stocking levels, and outdated records. Maintaining a strong, high-quality information system requires oversight and transparency of overall analytical outputs to reduce risks.

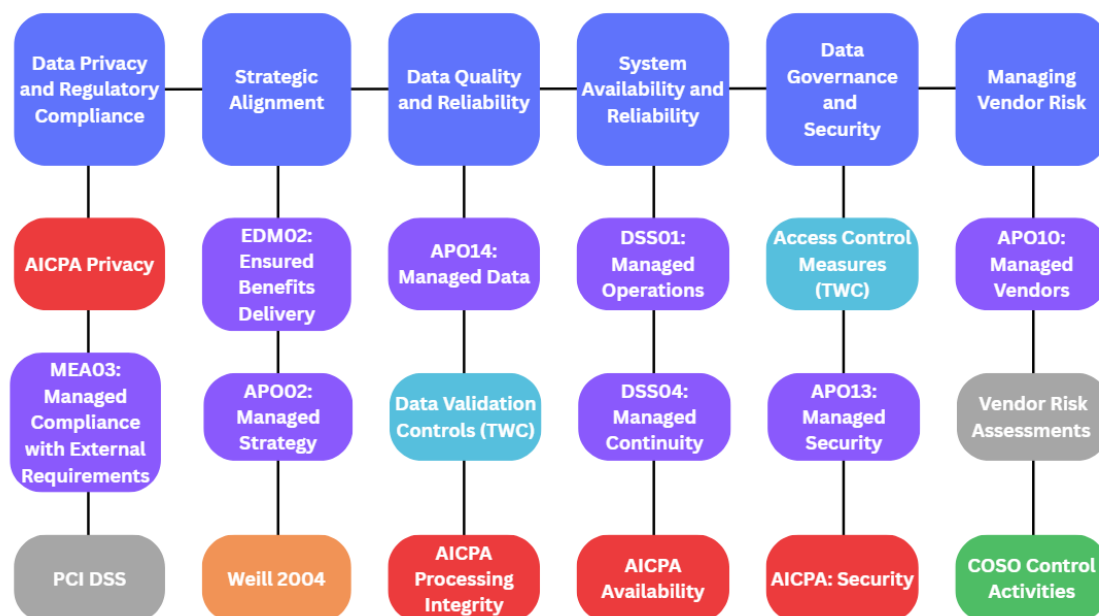
Overreliance on Analytics: Avoiding overreliance on analytical systems may lessen the risk of poor decision making and data integrity concerns. Analytics systems contain data that spans from very recent to historical, meaning data is stored in large volumes and varieties. This creates concern that systems will become flooded with inaccurate, untimely, or irrelevant data, which may be further used in the analysis and evaluation of critical solutions. However, supplementing analytical insights with human review, subjective information, and expertise curbs the possibility of low-quality data being used for decisions such as production schedules, stocking levels, vendor selection, and understanding customer needs. This also allows managers to train systems by validating and entering input into the system (Brau et al., 2024). Overall, big data is only as effective as the humans using it.

Integration Risk: Traditional supply chain systems have relied on managerial experience and human resources to drive performance and effectiveness. While big data fosters many new opportunities for implementation along the retail supply chain, firms have trouble integrating these new solutions with pre-existing systems. Many organizations have insufficient technological resources, a long learning curve for employees, and lack of commitment to using new technology in operations (Brau et al., 2024). Complexities in big data also create challenges in maintaining control and having resistance to change from management (Haessner et al., 2025). Therefore, governance must incentivize strategic use of big data and encourage entities along the supply chain to adopt its use.

Downtime/Unavailability: The most important component in using big data is ensuring systems are available readily, quickly, and continuously so that there is no threat of shutdown or unavailability (Lutfi et al., 2022). Retail supply chains rely on real-time data to support strategic activities and operations, so systems must be readily functioning. If a system shuts down or is unavailable, it could bring large costs to firms such as not supplying or sourcing enough material, misunderstanding customer needs, or facing risk of financial losses. Effective governance ensures backup data is secure and that restoration procedures are clearly intact so systems can continue to function.

Framework Model:

Figure 1: RBDA Governance Framework



The Retail Big Data and Analytics (RBDA) framework provides a high-level overview and guidance for how retail firms must implement effective IT governance for big data use. Pulling from existing frameworks and control guidelines such as ISACA's COBIT, AICPA Trust Service Criteria, COSO, and Weill (2004), as well as from established academic sources like Turner, Weickgenannt, & Copeland, the RBDA framework covers six domains: Data Privacy and Regulatory Compliance, Strategic Alignment of Big Data, Data Quality and Reliability, System Availability and Reliability, Data Governance and Security, and Managing Vendor Risk.

Following this framework, retail organizations can mitigate risks in big data use along their supply chains. Existing frameworks and principles integrated into the RBDA framework provide the foundation for risk mitigation, control implementation, and effective governance of big data use, specifically:

- COBIT 2019 – Governance processes for oversight and management of enterprise assets aimed to reduce risks, maintain compliance, and achieve business goals (COBIT, 2018)
- AICPA Trust Service Principles – Criteria for ensuring availability, security, processing integrity, confidentiality, and privacy in organizational IT systems (AICPA, 2022)
- COSO – Framework outlining effective and efficient processes for risk mitigation and internal control for business processes (COSO, 2013)
- Weill – Establishes a framework for allocating how to use assets (decision rights) and how to hold parties accountable for decisions (Weill, 2004)

A detailed description of each domain is as follows:

Data Privacy and Regulatory Compliance:

Maintaining data privacy protects sensitive consumer data from misuse and harm. Sensitive consumer information, such as credit card information, is processed frequently in retail supply chains. Therefore, effective governance mandates oversight to ensure sensitive data is handled properly and complies with requirements. Established frameworks, such as MEA03: Managed Compliance with External Requirements, assist firms in managing compliance to standards, such as the Payment Card Industry Data Security Standard in retail settings, securing sensitive payment card information (COBIT, 2018; PCI Security Standards Council, n.d.). Additionally, by understanding the Privacy Principle from AICPA, retail firms can develop proper controls to secure access (AICPA, 2022). Our framework encourages implementing privacy management measures such as privacy policies, ongoing monitoring of compliance, and encryption to maintain consumer trust.

Strategic Alignment of Big Data:

Big data must be used to advance, not impede, business goals and objectives. However, risks exist in the multiple parties involved in retail supply chains, as data can be used, stored, or

processed in a way that satisfies individual goals, but not organizational goals. There is no established authority over data ownership, so governance must monitor whether data delivers optimal value for the organization, aligning with strategy and holding entities accountable for data use. COBIT's EDM02: Ensured Benefits Delivery and APO02: Managed Strategy ensure governance processes maximize organizational value and align with retail supply chain objectives (COBIT, 2018). This domain also incorporates research from Peter Weill (2004), highlighting the importance of allocating rights to big data use to the correct people. Understanding this domain mitigates risk of data misuse and encourages incentives to properly use big data for pricing optimization, supply chain efficiency, and demand forecasting.

Data Quality and Reliability:

Decisions made along retail supply chains are dependent on high quality, relevant data. Information should be accurate and complete, to avoid erroneous mistakes such as production failures, inventory stockouts, and logistics issues. This domain emphasizes the importance of developing controls to ensure data quality and performance throughout its use, while mitigating bias. APO14: Managed Data lays the foundation for how data must be effectively managed to maintain quality in business use (COBIT, 2018). Further, our framework provides guidance on selecting data validation controls such as field checks, limit checks, and validity checks, to protect data quality (Turner, Weickgenannt, & Copeland, 2017, pp. 135-137). The Processing Integrity Principle from AICPA also supports reliable data processing controls (AICPA, 2022).

System Availability and Reliability:

Outdated, incomplete, or incorrect information is costly, and leads to suboptimal decisions being made. When systems malfunction, risk of downtime may cause information to be untimely, and retailers rely heavily on real-time information with big data like POS systems. This domain allows firms to manage systems and implement mechanisms to ensure data is readily available and reliable. Pulling from DSS01: Managed Operations, DSS04: Managed Continuity, and AICPA's Availability Principle, understanding this domain allows firms to implement effective measures to ensure systems run continuously and reliably (COBIT, 2018; AICPA, 2022).

Data Governance and Security:

Maintaining secure systems begins with identifying who is responsible for securing data and limiting access to those who are authorized. A proper governance structure must address accountability and authority, ensuring that sensitive data is accessed only by those permitted. Also, vulnerabilities to cybersecurity structures can lead to costly loss of operational efficiency. Therefore, this domain establishes the need for access management and accountability to keep data secure and well-kept. Incorporating perspectives from [APO13: Managed Security](#) and [AICPA's Security Principle](#) can help retailers maintain definition of authority and control unauthorized access through mechanisms such as policies, authority tables, and access control measures (COBIT, 2018; AICPA, 2022; Turner et al., 2017, pp. 106-113).

Managing Vendor Risk:

Retailers depend on third parties for big data storage, logistics, and processing. Dependence on external vendors introduces the risk of improper use of big data or security failures if vendors do not have the proper controls or are not assessed. Conducting thorough vendor assessments is a strong preventative measure that can be used to ensure vendors are adequately trusted and not likely to create backlogs in operations. In addition, [COSO's internal control activities](#), specifically authorization of transactions and adequate records, holds vendors accountable for risk (COSO, 2013). [APO10: Managed Vendors](#) further outlines how to properly manage vendor selection and mitigate third-party related risks (COBIT, 2018).

Collectively, these six domains form a comprehensive governance guideline and overview that provides a best-practice approach for overseeing big data use in retail supply chains. Each domain reinforces the others – for example, system availability can impact data quality and reliability, demonstrating the necessity of an integrated framework. Additionally, the framework maps each identified risk into the domain structure to assist in prioritizing controls and mitigation activities. By aligning these domains, retailers can design proper risk management procedures to implement effective controls, enhance decision-making, and optimize efficient big data-driven supply chain functions.

Conclusion:

As retail supply chains become more reliant on big data, strong IT governance will become an essential aspect of business operations. Reliance on big data improves decision-making and efficiency but also opens the door to a number of cybersecurity, compliance, and organizational risks. Without clear guidance and oversight, companies are at risk of mismanaging or misinterpreting sensitive data and making decisions that don't align with company goals. Because of these threats, retail firms need a structured framework and approach to managing how big data is used across their supply chains.

Adopting a formal governance framework, such as RBDA, will provide significant benefits to retail organizations. RBDA mitigates the previously identified risks through the six domains: Data Privacy and Regulatory Compliance, Strategic Alignment of Big Data, Data Quality and Reliability, System Availability and Reliability, Data Governance and Security, and Managing Vendor Risk. By following the domains within this framework, retail organizations will ensure that their data is accurate, secure, and aligned with business strategy, while still gaining the increased efficiency and effectiveness of analytics.

A well-defined governance model strengthens stakeholder confidence by proving that a firm can manage data and security risk, while also capitalizing on the benefits of big data usage. As retail supply chains grow more advanced and more reliant on big data analytics, organizations that prioritize governance will create a strategic competitive advantage and will be the most capable of turning big data into sustainable value.

Annotated Bibliography

AICPA. (2022). 2017 Trust Services Criteria for Security, Availability, Processing Integrity, Confidentiality, and Privacy (with Revised Points of Focus - 2022). AICPA.

Abstract: “ASEC has developed a set of criteria (trust services criteria) to be used when evaluating the suitability of the design and operating effectiveness of controls relevant to the security, availability, or processing integrity of information and systems used to provide products or services, or the confidentiality or privacy of information processed by the systems used to provide products or services at an entity, a division, or an operating unit of an entity. In addition, the trust services criteria may be used when evaluating the design and operating effectiveness of controls relevant to the security, availability, processing integrity, confidentiality or privacy of a particular type of information processed by one or more of an entity’s systems or one or more systems used to support a particular function within the entity. This document presents the trust services criteria.”

Usage: We will use this information from AICPA Trust Services Principles to analyze an outside framework to reference when creating our framework. This framework is the criteria for ensuring availability, security, processing integrity, confidentiality, and privacy in organizational IT systems. Alongside the other existing frameworks, AICPA will allow us to create our framework with controls in mind for retail focused companies and their supply chains.

Ardhaninggar, E. A., & Ramli, K. (2024). A review of cybersecurity framework implementation for retail industry: Challenges and recommendations. *ARRUS Journal of Engineering and Technology*, 4(2), 212-225. <https://doi.org/10.35877/jetech3434>

Abstract: “Efforts to improve services by opening online data access to customers have become straightforward targets for cybercrime. Unfortunately, the researchers' findings reveal that there is presently no cybersecurity framework that completely aligns with the fundamental principles of the retail industry. This paper then analyzes the deployment of cybersecurity frameworks across

various industrial sectors to determine which framework—or combination of frameworks—best aligns with the fundamental values of the retail industry. We compare the essential points of each cybersecurity framework with the TWOS Matrix, which represents the core values of the retail industry. We suggest NIST CSF, ISO/IEC 27001:2022, and Essential Eight as the best combination of cybersecurity frameworks for the retail industry. Therefore, the comparative analysis results recommend RCF as a novel framework suitable for implementation in the retail industry”

Usage: This article is unique to our research because it identifies the absence of a cybersecurity focused framework that is designed specifically for the special needs of the retail industry. We plan to use this source to assist in establishing our new framework. This article emphasizes efficiency and integrity with data management and security frameworks, which are both factors we plan to include in our analysis. We may tie this article back to our initial risks section to create a better foundation for our framework.

Bakshi, S. (2016). Performance measurement metrics for IT governance. *ISACA Journal*, 6, 21-27.

Abstract: During the past 30 years, enterprises have been embracing new methods to transform their operations to use IT and related technology to provide a higher level of customer service. The pace at which enterprises are adopting these new methods is rapid. To handle the speed of this transformation, management relies on technology resources and vendors, resulting in an increased dependency on technology and skilled resources. The pace and dependencies can create a lack of enterprise control; therefore, enterprises use key performance indicators (KPIs) to measure the performance of IT service delivery.

Usage: This article and the conclusion the authors provide will be used in our report to help create our working definition of IT Governance. We specifically will use this to reference how to

manage stakeholder needs in our definition, and the importance of this metric. Lastly, the article mentions vendor relationships in regards to technological resources.

Brau, R. I., Sanders, N. R., Aloysius, J., & Williams, D. (2024). The role of the human factor in decision making in digitalized retail supply chains. *Journal of Business Logistics*, 45(1). <https://rave.ohiolink.edu/ejournals/article/433310876>

Abstract: “Our research reveals the continued and evolving role of the human factor in decision making in digitalized retail supply chains. We compare managerial roles in a pre- and post-COVID era through conducting in-depth interviews of 25 executives spanning the retail supply chain ecosystem. We use grounded theory to develop four main contributions. *First*, we find that the involvement of managerial judgment is found to be progressively greater moving up the retail supply chain, away from the customer and the demand signal. *Second*, integration of analytics and judgment is now the primary method of decision making, and we identify elements needed for success. *Third*, we develop an essential framework for a successful integration process. *Fourth*, we isolate the necessary components of a successful process for analytics/AI implementation. Our paper offers important insights into how analytics and AI are—and should be used—in judgment and decision making and opportunities for researchers to understand the changing role of the human factor in digitalized retail supply chains.”

Usage: This article discusses how the relevance of AI and big data management are being addressed by company management in retail supply chains. Based on the research and interviews conducted with 25 industry executives, the authors conclude that, although data tracking is a profound part of a company's daily processes, human judgement is still massively crucial. We plan to use this source in our report to bridge the gap between the risks of big data in retail supply chains, and the creation of our new framework. Specifically, we plan to use the findings on integration and human-specific data risks to establish a need for employee accountability in our new framework.

COBIT. (2018). COBIT ® 2019 Framework: Governance and Management Objectives, ISACA, Schaumburg, IL.

Abstract: “Over the years, best-practice frameworks have been developed and promoted to assist in the process of understanding, designing and implementing enterprise governance of IT (EGIT). COBIT® 2019 builds on and integrates more than 25 years of development in this field, not only incorporating new insights from science, but also operationalizing these insights as practice.

From its foundation in the IT audit community, COBIT® has developed into a broader and more comprehensive information and technology (I&T) governance and management framework and continues to establish itself as a generally accepted framework for I&T governance.”

Usage: We use this resource to help assist in our creation of a new framework. We will use this work to better understand the purpose and usage of existing frameworks, to establish our specific retail supply chain orientation. This COBIT framework is used to reference governance processes for oversight and management of enterprise assets.

COSO. (2013). Internal Control - Integrated Framework: Executive Summary. COSO.

Abstract: “Internal control helps entities achieve important objectives and sustain and improve performance. COSO’s Internal Control—Integrated Framework (Framework) enables organizations to effectively and efficiently develop systems of internal control that adapt to changing business and operating environments, mitigate risks to acceptable levels, and support sound decision making and governance of the organization. Designing and implementing an effective system of internal control can be challenging; operating that system effectively and efficiently every day can be daunting. New and rapidly changing business models, greater use and dependence on technology, increasing regulatory requirements and scrutiny, globalization, and other challenges demand any system of internal control to be agile in adapting to changes in business, operating and regulatory environments...”

Usage: We use the COSO framework to outline effective and efficient processes for risk mitigation and internal control for business processes. This framework will be referenced and used throughout our framework design to address the previously mentioned purposes. COSO will be particularly useful for managing vendor risks in our framework.

Curtis, B. (2020, June 29). *The value of IT governance*. ISACA.

<https://www.isaca.org/resources/news-and-trends/industry-news/2020/the-value-of-it-governance>

Abstract: “Many organizations misunderstand IT governance's purpose and value because few qualified professionals are adept in IT governance. However, governance and assurance professionals can change this perception via strategic alignment and the effective use of governance enablers such as principles, processes and policies. Additionally, via the employment of information and technology, enterprises can tailor the governance of enterprise IT (GEIT) framework to optimize their people, skills and competencies and improve their culture, behavior and ethics. By aligning assurance functions such as risk management, IT audit and cyber, organizations can create strong governance. Assurance helps enterprises create plans aimed toward optimizing business goals, establishing alignment and reducing risk to enterprise objectives.”

Usage: We will use this article in our working definition of IT Governance. Specifically, the author discusses how IT governance is an organizational mechanism that encourages “the ability for IT to achieve business goals”. This quote will play into how we define IT governance, and why it is so crucial for the emphasis of setting and achieving business goals.

Haessner, P., Haessner, J., & Thomas, J. (2025). Maximizing retail potential: The role of big data analytics. *Journal of Strategic Innovation and Sustainability*, 19(4), 67-82.

<https://doi.org/10.33423/jsis.v19i4.7486>

Abstract: "The retail industry has been transformed by technological advancements and evolving consumer behavior. This paper examines the benefits, challenges, and strategic implications of Big Data Analytics (BDA) in the retail sector through a comprehensive review of recent literature. The findings highlight three key opportunities for leveraging BDA in retail: enhancing customer relationships, improving operational efficiency, and gaining strategic advantages. However, several challenges persist, including privacy and security concerns, inadequate data-driven support systems, complexities in data visualization and processing, and insufficient governmental support. In addition to exploring the potential advantages and limitations, this study emphasizes the strategic role of BDA in facilitating data-driven decision-making for retailers. By balancing these opportunities and challenges, retail firms can harness Big Data as a critical source of competitive advantage. Finally, the paper identifies future research directions to further advance the application of Big Data in the retail sector, aiming to address existing gaps and unlock its full potential."

Usage: In this article, the authors explore how the retail industry has been transformed through Big Data Analytics (BDA) and a shift in how companies handle consumer behavior, data, and technology. We plan to use this study in our report to better understand the shift towards digital business models, and the importance of using a clear, strategic framework when implementing new data practices. We also intend to emphasize the importance of privacy and security concerns when managing big data across international supply chains.

Johnson, O., Brown, W., & Wilson, G. (2024). The role of big data analytics in retail marketing and supply chain optimization. <https://doi.org/10.20944/preprints202407.2058.v1>. (pp. 5-11)

Abstract: "This research explores the pivotal role of big data analytics in enhancing retail marketing and supply chain optimization. As the retail industry undergoes rapid digital transformation, big data analytics has emerged as a critical tool for gaining competitive advantage

and improving operational efficiency. The study delves into the various applications of big data analytics, including customer segmentation, personalized marketing, demand forecasting, inventory management, dynamic pricing, and supply chain optimization. It highlights how retailers leverage data from diverse sources to gain insights into consumer behavior, market trends, and operational performance, leading to more informed decision-making. The findings underscore the significant benefits of big data analytics in enhancing customer experiences through targeted engagement and personalized service. Additionally, the research addresses the ethical considerations and challenges associated with data privacy, security, and potential biases in analytics models. It emphasizes the importance of maintaining high standards of data quality and ethical practices to ensure the responsible use of data. The conclusion reiterates the transformative impact of big data analytics in the retail sector, suggesting that while it offers substantial advantages, careful management of its challenges is crucial. The study provides insights into how retailers can harness the power of big data analytics to drive growth, innovation, and long-term success in an increasingly competitive and data-driven market environment.”

Usage: This article explores the impact that big data has had on the retail industry in recent years. Although big data has created clear advantages for retailers wanting to reach and connect with consumers, the authors also explore the negative consequences that big data management can have, not limited to privacy, cybersecurity, and ethical concerns. We plan to use this article throughout our report because of the connections drawn between positive and negative consequences, and how to properly implement data protection and management frameworks to protect consumer and company data. One main callout from the article is the emphasis of maintaining high standards for data quality and privacy, which is something we plan to tie into our new framework.

Lutfi, A., Alrawad, M., Alsyouf, A., Almaiah, M. A., Al-Khasawneh, A., Al-Khasawneh, A. L., ... & Ibrahim, N. (2022). Drivers and impact of big data analytic adoption in the retail industry: A quantitative investigation applying structural equation modeling. *Journal of Retailing and Consumer Services*, 70, 103129.

<https://doi.org/10.1016/j.jretconser.2022.103129>

Abstract: “Big data analytics (BDA) adoption has gained attention in both practical and theoretical circles owing to the opportunities and advantages that can be reaped from it. In theory, the majority of researchers have evidenced the benefits of BDA, although barriers to its adoption have also been mentioned. This study draws upon the technology-organisation-environment framework and resource-based view theory to propose an integrated model that examines the drivers and impact of BDA adoption in the retail industry in Jordan. The proposed single model encapsulates the aspects of BDA adoption and performance. The study makes use of an online questionnaire survey to collect the required data, and the research model is eventually validated based on 132 responses gathered from the retail industry in Jordan. The findings highlight two major observations. The first is that relative advantage, organisational readiness, top management support, government support, data variety and data velocity all have a significant influence over BDA adoption. The second observation is that a significant association exists between BDA adoption and firm performance, providing information on the way firms can enhance their BDA adoption for enhanced performance. This study contributes to literature dedicated to examining BDA in terms of its drivers and impact on performance and can be used as a reference in developing nations.”

Usage: We plan to use the findings of this study because the context for how companies can best implement BDA, and why strategic alignment is so crucial. We also want to tie in their ideas of data velocity and how that can increase data security to help mitigate key risks.

PCI Security Standards Council. (n.d.). *PCI Data Security Standard (PCI DSS)*.

<https://www.pcisecuritystandards.org/standards/pci-dss/>

Abstract: “PCI DSS was developed to encourage and enhance payment card account data security and facilitate the broad adoption of consistent data security measures globally. PCI DSS provides a baseline of technical and operational requirements designed to protect payment account data.”

Usage: We plan to use this article to assist in establishing our framework, specifically in the privacy domain. Along with the other frameworks we’re referencing, this article helps establish our foundational understanding of the frameworks and best practices.

Niu, Y., Ying, L., Yang, J., Bao, M., & Sivaparthipan, C. B. (2021). Organizational business intelligence and decision making using big data analytics. *Information Processing & Management*, 58(6). <https://doi.org/10.1016/j.ipm.2021.102725>

Abstract: This article explains how massive amounts of unstructured, unorganized data can be transformed into “actionable information” through Business Intelligence (BI), and how this transformation of data can improve business productivity. The authors identified major hurdles found in current governance plans and created a new framework called Optimized Data Management (ODM-BDA). This new framework addresses many of their previous concerns and is based in cloud technology used to improve implementation and organization of systems, especially in relation to financial data.

Usage: We plan to use this article in our report by referencing their process for identifying data privacy risks, and create a new framework to address those identified issues. From our paper’s perspective as an IT auditor, this source will be crucial as it provides a roadmap for quality assurance and how to monitor error rates for improved system and framework reliability.

Turner, L., A.Weickgenannt, & M.K.Copeland (2017), Accounting Information Systems: Controls and Processes, 3rd ed., Wiley

Abstract: This is the textbook for ISA/ACC305, and provides a basic overview of important definitions, concepts, and frameworks to prepare students for their studies.

Usage: We will use the TWC text to establish domains of our framework, and will specifically reference Data Validation Controls and Access Control Measures. These specific controls fall into our domains pertaining to Data Quality and Reliability and Data Governance and Security. Along with the other established frameworks, we will use the TWC text as a guideline for definitions, controls, and functionality of frameworks and their domains.

Weill, P. (2004). Don't just lead, govern: How top-performing firms govern IT. MIS Quarterly Executive, 3(1), 1-17.

Abstract: "Top-performing enterprises succeed in obtaining value from IT where others fail, in part, by implementing effective IT governance to support their strategies and institutionalize good practices. IT governance involves specifying decision rights and accountabilities for important IT decisions. The goal is to encourage "desirable behaviors" in the use of IT. In studying the IT governance of more than 250 enterprises in 23 countries, we found a wide array of IT governance arrangements. Enterprises assign "decision rights" to different "archetypes" (Business or IT Monarchy, Federal, Duopoly, Feudal, or Anarchy) to govern five key IT decisions (IT investment, architecture, principles, application needs, and infrastructure). Top-performing enterprises govern IT differently from each other and from average enterprises. Firms leading on growth decentralize more of their IT decision rights and place IT capabilities in the business units. Those leading on profit centralize more decision rights; senior business leaders make the major IT decisions. Top performers design their IT governance to reinforce their performance goals and link IT governance to the governance of their other key enterprise assets and desired behaviors. A case

study of State Street Corporation illustrates IT governance evolution and a method to diagrammatically represent IT governance.”

Usage: The Weill research and article will be used in our report to help establish our working definition of IT governance. Additionally, this framework from Weill will be used to establish our new framework. This framework discusses the importance of understanding decision rights and accountability in data management, and that will go towards our definition of IT governance. The Weill framework is a guideline for how to properly use assets and establish decision rights.